

Machine Learning Algorithms Laboratory

Experiment 1 Report

Experiment No.	1
Experiment Title	Working with Python Packages - NumPy, Pandas, SciPy, Scikit-learn and Matplotlib 1
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Submission Date	

1 Objective

Becoming familiar with commonly used Python libraries used in machine learning. Public datasets were explored and basic Exploratory Data Analysis was performed to understand the data before applying machine learning models.

2 Problem Statement

Different datasets were explored using Python libraries such as NumPy, Pandas and Matplotlib. Basic statistics and visualizations were generated to study the data distribution, identify missing values and observe relationships between features. The suitable machine learning task for each dataset was also identified.

3 Dataset Description

Dataset	Description
Dataset Name	Iris Dataset
Dataset Source	Kaggle
Number of Samples	150
Number of Features	4 Numerical Features
Number of Classes	3 (Setosa, Versicolor and Virginica)
Missing Values	No Missing Values
Purpose	Exploring data using statistical summary and visualizations.

Dataset	Description
Dataset Name	Loan Prediction Dataset
Dataset Source	Kaggle
Number of Samples	614

Number of Features	13 Columns
Target Variable	Loan Status
Missing Values	Present in a few columns
Purpose	Understanding feature distribution and relationships using EDA.

Dataset	Description
Dataset Name	Email Spam Dataset
Dataset Source	Kaggle
Number of Samples	5573
Number of Features	2 Columns
Target Variable	Category (Ham/Spam)
Missing Values	No Missing Values
Purpose	Exploring the class distribution of spam and ham emails using Exploratory Data Analysis.

Dataset	Description
Dataset Name	Diabetes Prediction Dataset
Dataset Source	Kaggle
Number of Features	Medical and Demographic Features
Target Variable	Diabetes
Missing Values	No Missing Values
Purpose	Studying the distribution of diabetes cases and identifying important numerical features using Exploratory Data Analysis.

4 Exploratory Data Analysis

4.1 Iris Dataset

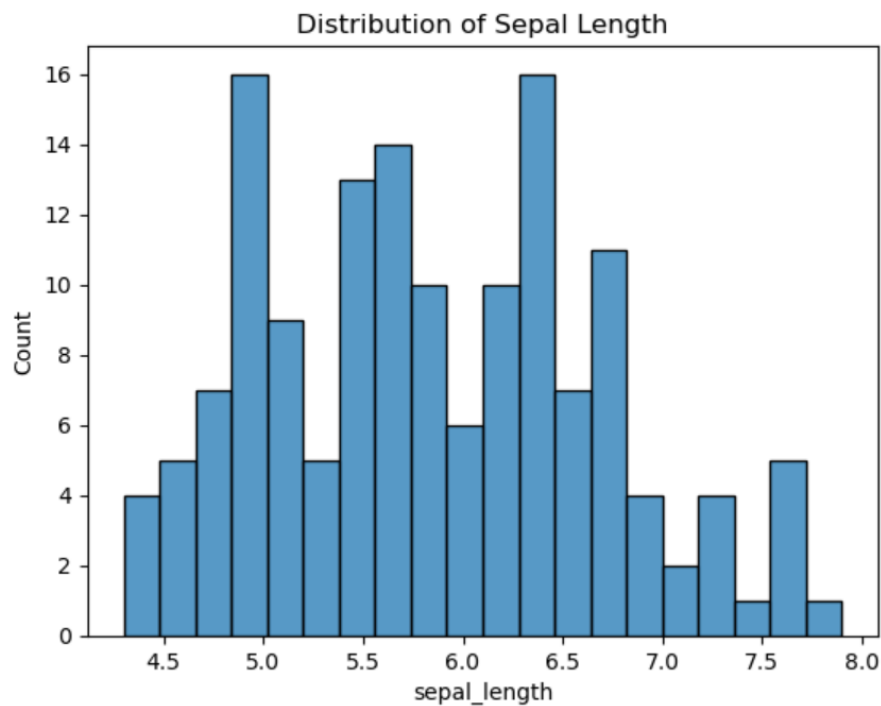


Figure 1: Histogram of Sepal Length

The histogram shows that most sepal length values are concentrated around the middle range.

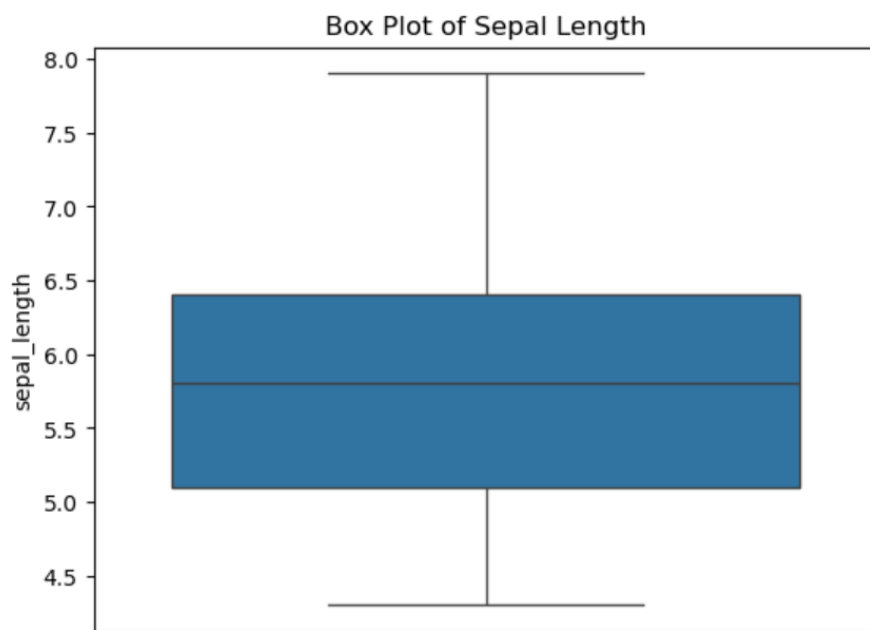


Figure 2: Box Plot of Sepal Length

The box plot shows the spread of the data and indicates a few values outside the normal range.

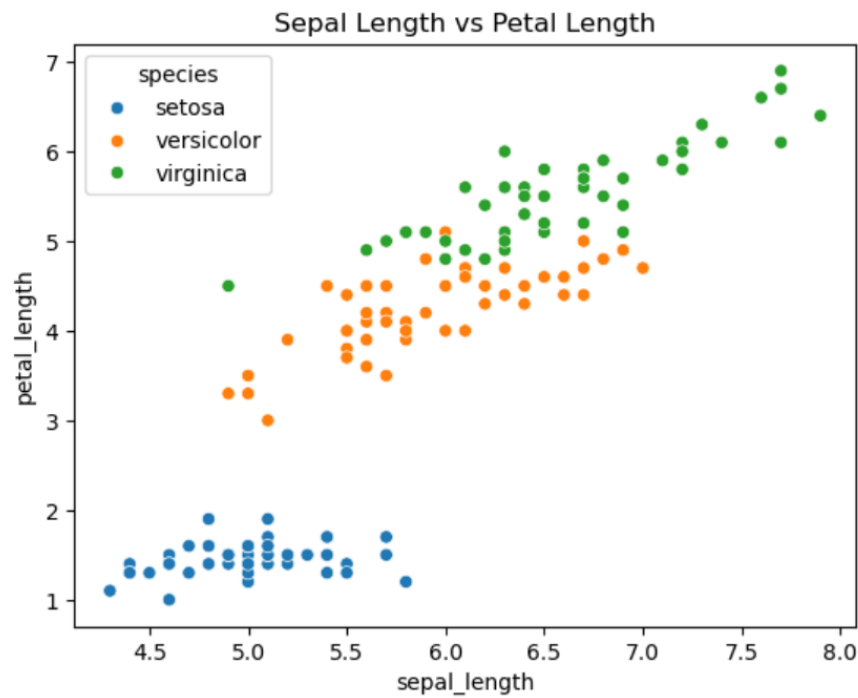


Figure 3: Scatter Plot of Sepal Length vs Petal Length

Some flower classes form separate groups while a small overlap is also visible.

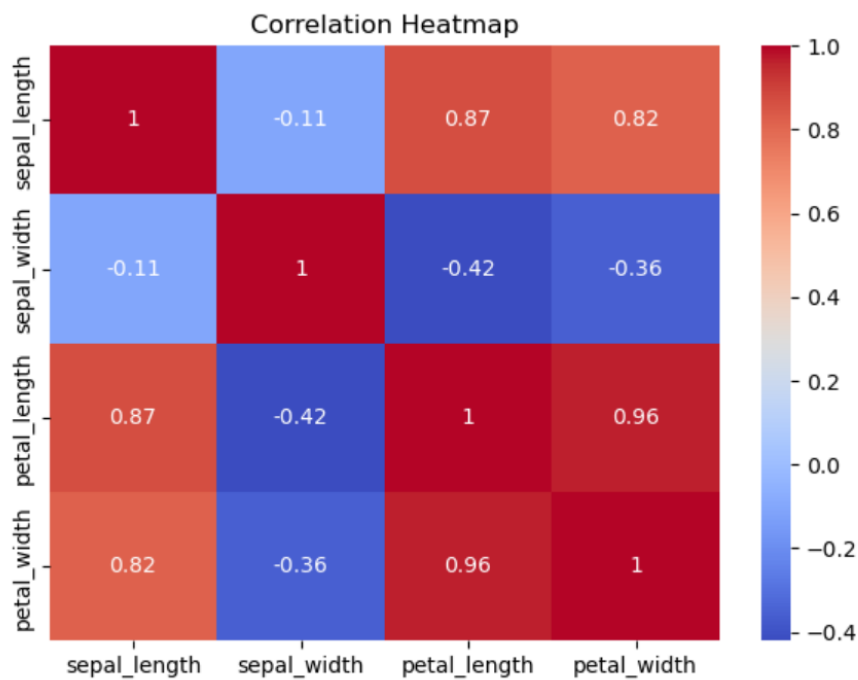


Figure 4: Correlation Heatmap of Iris Dataset

4.2 Loan Prediction Dataset

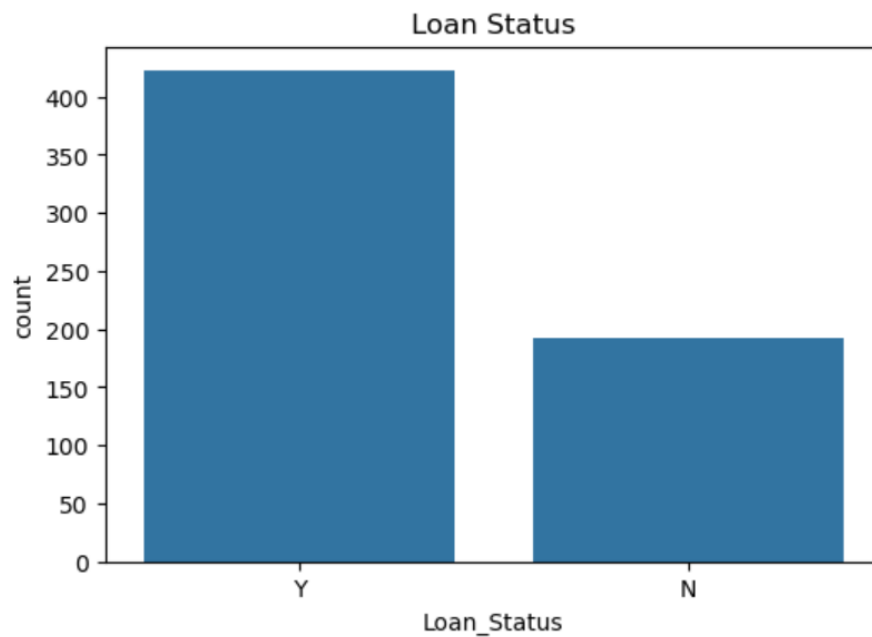


Figure 5: Loan Status Distribution

The graph shows the number of approved and rejected loan applications.

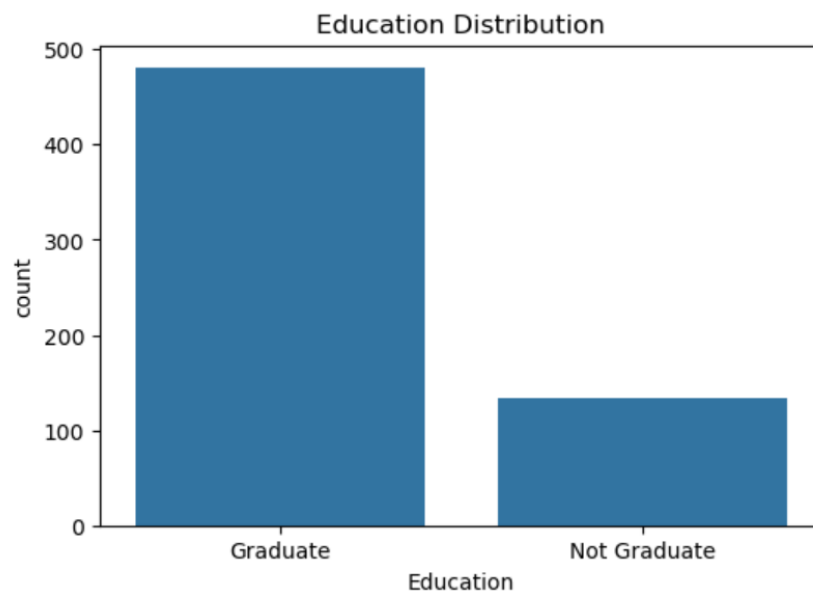


Figure 6: Education Distribution

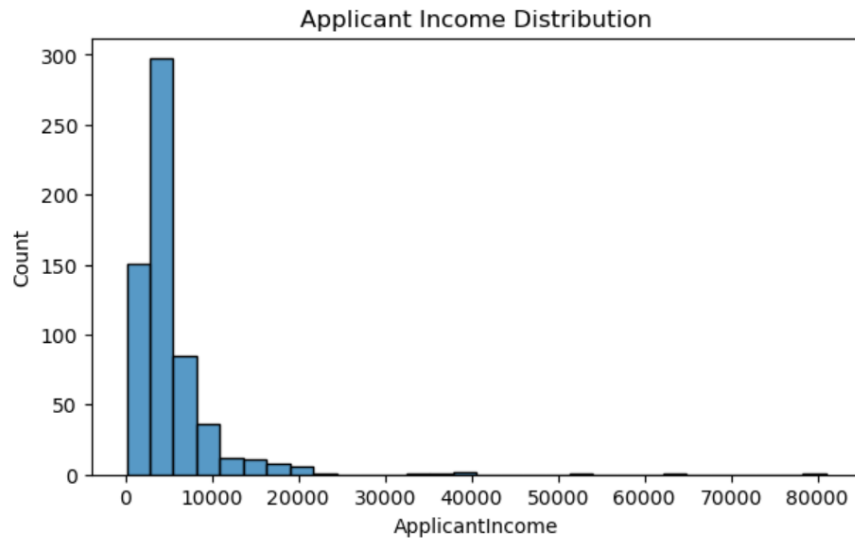


Figure 7: Distribution of Applicant Income

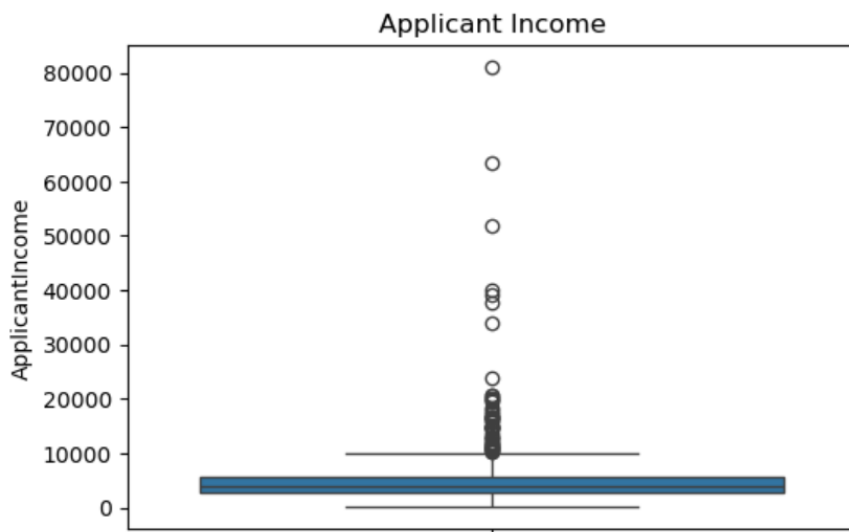


Figure 8: Box Plot of Applicant Income

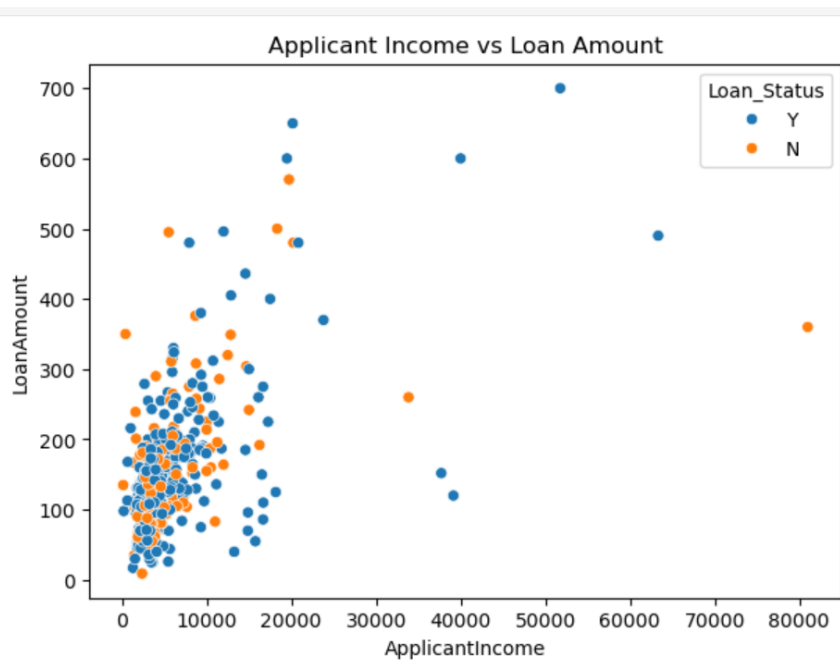


Figure 9: Applicant Income vs Loan Amount

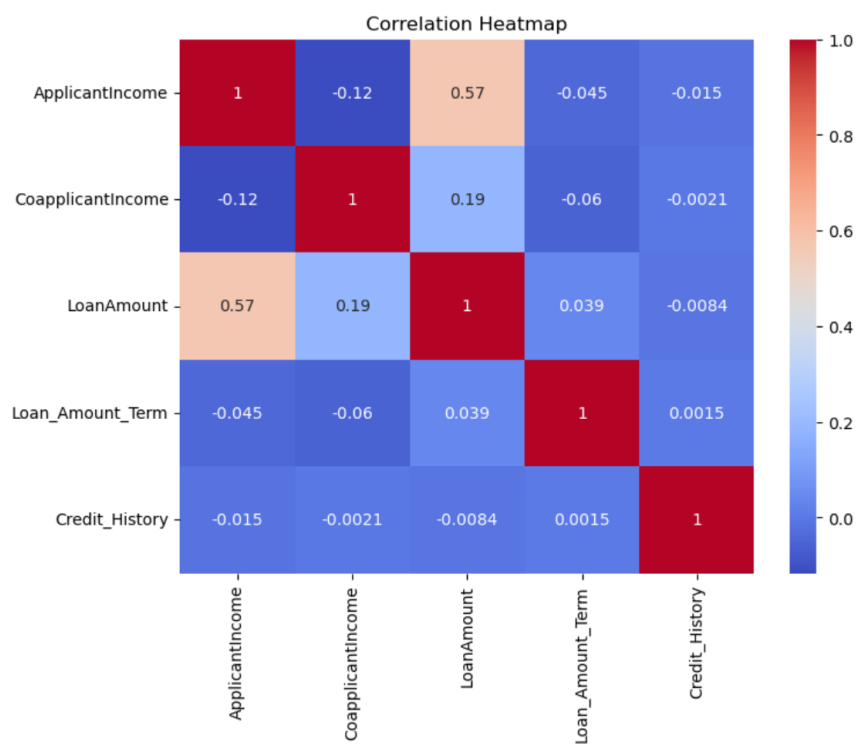


Figure 10: Correlation Heatmap of Loan Dataset

4.3 Email Spam Dataset

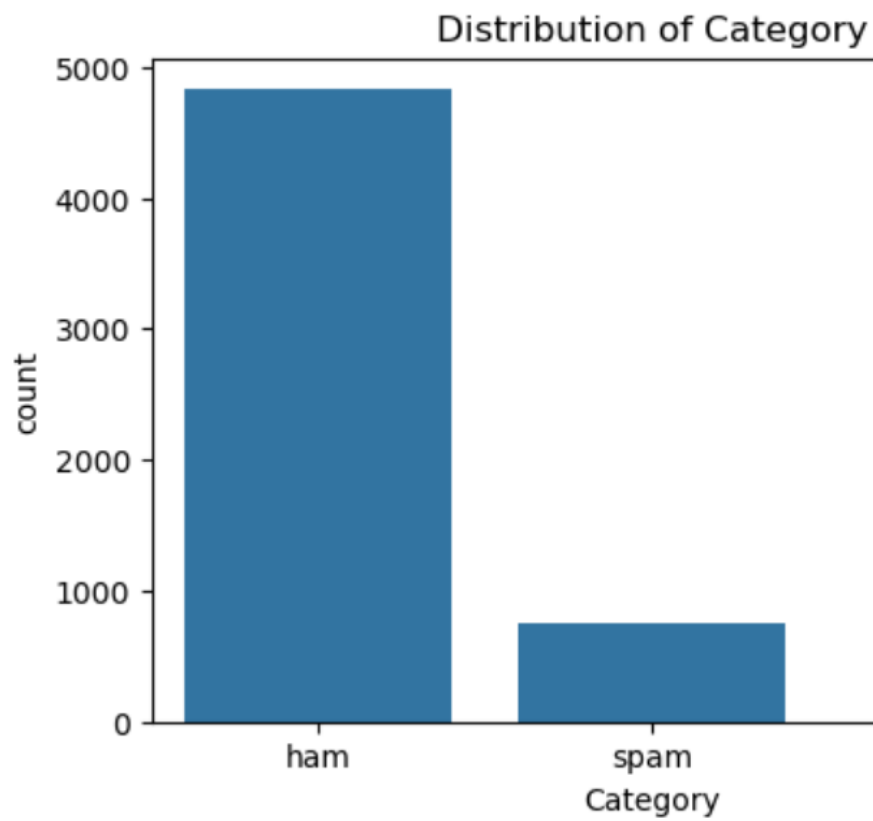


Figure 11: Distribution of Email Categories

The bar graph shows the distribution of ham and spam emails in the dataset.

4.4 Diabetes Dataset

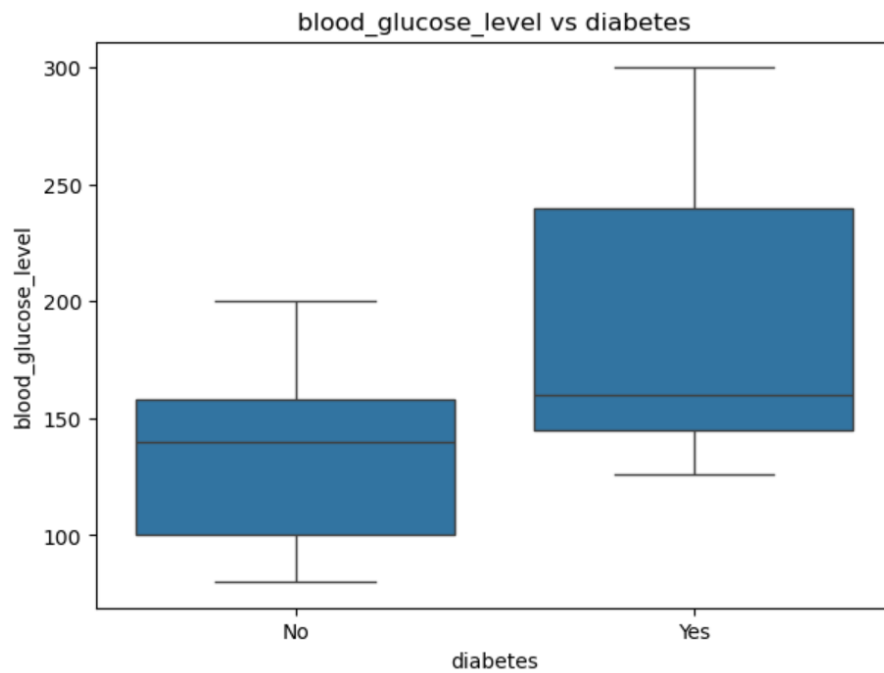


Figure 12: Blood Glucose Level vs Diabetes

Individuals diagnosed with diabetes generally have higher blood glucose levels than those without diabetes.

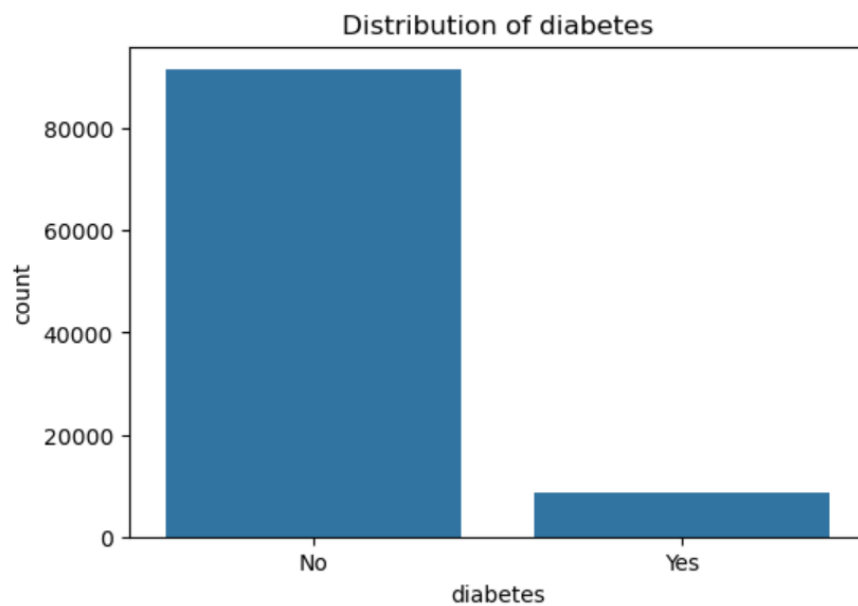


Figure 13: Distribution of Diabetes Cases

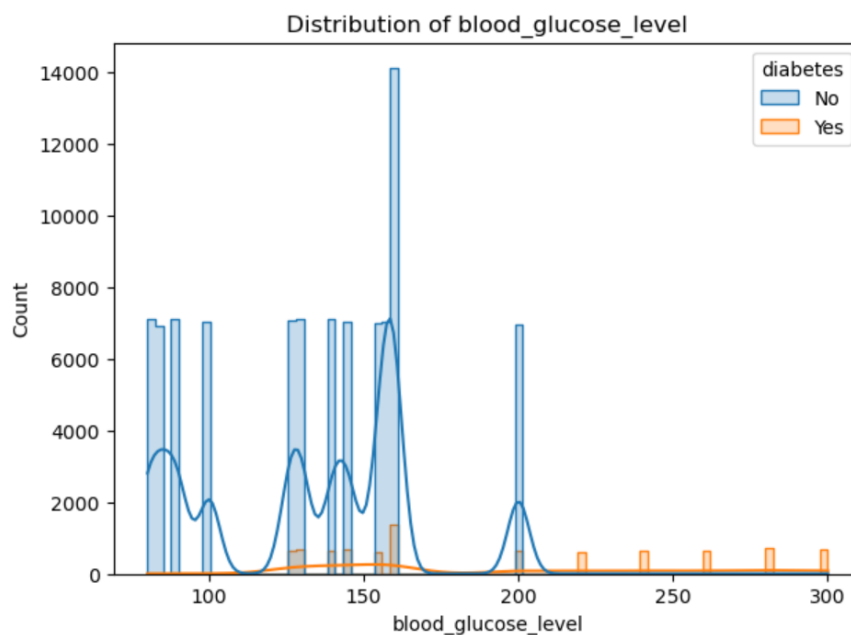


Figure 14: Distribution of Blood Glucose Levels

5 Data Preprocessing

The datasets were inspected using Pandas. Missing values were checked. Aim of this experiment was Exploratory Data Analysis, So no additional preprocessing was done.

6 Discussion

This experiment provided a basic understanding of working with machine learning datasets using Python libraries. Different plots helped in studying feature distributions, identifying outliers and understanding the relationship between variables. It also highlighted the importance of EDA.

7 Conclusion

In this experiment datasets were explored using Python libraries such as Pandas and Matplotlib. Exploratory Data Analysis was performed using summary statistics and different visualizations. The experiment helped in understanding the datasets and the initial steps involved in a machine learning workflow.

8 References

1. NumPy Documentation. <https://numpy.org/doc/>
2. Pandas Documentation. <https://pandas.pydata.org/docs/>
3. Scikit-learn Documentation. <https://scikit-learn.org/stable/>

4. Kaggle Datasets. <https://www.kaggle.com/datasets>
5. UCI Machine Learning Repository. <https://archive.ics.uci.edu/>